

Appendix

Perception Entropy for Path Planning in Autonomous Rover Navigation

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A. Dataset

We use a subset of the ROVER dataset [1], a multi-season outdoor visual SLAM benchmark collected with a small-scale UGV equipped with an Intel RealSense D435i RGB-D camera, where the calibrated and synchronized sensor suite provides millimetre-accurate ground-truth poses in TUM format aligned to a global frame. For our experiments, we use the ROVER_data/realsense_D435i/sequence, which contains paired color images (rgb/) and 16-bit PNG depth maps (depth/), converting raw depth values to metres with a scale factor of 10^{-3} and associating RGB-depth pairs via timestamps from rgb.txt and depth.txt with a maximum 30 ms difference to ensure temporal alignment. Ground-truth camera poses are stored in groundtruth.txt in the standard TUM RGB-D pose representation $[x, y, z, q_x, q_y, q_z, q_w]$, and we follow the ROVER protocol by discarding frames 1–39 to allow pose convergence, using frames 40–230 as our training window and selecting the pose at frame 60 as the navigation goal for downstream planning and control tasks.

B. Occupancy Grid Construction and Fog-of-War Setup

We construct a 2-D occupancy grid on the ground plane using the (x, y) components of the ground-truth robot poses from frames [40, 230]. The grid domain is defined by computing an axis-aligned bounding box over all valid poses in this range and expanding it by 1.0 m on each side to cover the full navigable area, discretised at resolution $r = 0.05$ m.

For each frame i , the 16-bit depth image is back-projected into 3-D points in the camera frame using pinhole intrinsics ($f_x = f_y = 610$ px, $c_x = 320$ px, $c_y = 240$ px), clipped to the range [0.5, 10.0] m to reject sensor noise. These points are transformed into world coordinates using the ground-truth position (x_i, y_i) and heading $\psi_i = \text{atan2}(\Delta y_i, \Delta x_i)$ derived from consecutive pose differences. World points are then projected onto the 2-D grid: a cell is marked *occupied* ($= 1$) if the fraction of points within the obstacle height band [0.8 m, 2.5 m] exceeds a hit-ratio threshold of 0.2 (with a minimum of 3 depth hits per cell); otherwise, it is marked *free* ($= 0$), or remains *unknown* (-1) if unobserved.

Observations from all frames are fused via confidence-weighted accumulation with per-cell confidence $\tanh(\text{ray count}/20)$; a new observation overwrites an

existing one only if it has strictly higher confidence, and high-confidence free space (confidence ≥ 0.3) can override a previous obstacle label. The resulting global occupancy grid ("god-map") is precomputed once, kept fixed during training as a hidden oracle for fog-of-war simulation, and at each time step the RL agent observes only a local, ego-centric crop of this grid with fog-of-war applied, yielding a compact top-down view of nearby free space and obstacles.

C. Active Perception

The agent operates under a fog-of-war constraint: the local map $\mathcal{M}$ is initialised as entirely unknown (-1 everywhere) and is revealed incrementally through scanning. Prior to training, a single 60° forward cone raycast is performed at the agent's starting position (facing North), revealing cells from the oracle map into $\mathcal{M}$. During an episode, a scan is triggered before each step if: $d_{\text{dir}}(s, a) < \tau_{\text{scan}}$ and the ray in direction a terminates at an unknown cell ($\text{stop_type}(s, a) = -1$), where $\tau_{\text{scan}} \in \{5, 10, 15, 20\}$ cells is the scan depth threshold, $d_{\text{dir}}(s, a)$ is the directional distance to the nearest unknown cell ahead, and s is the current cell. Each triggered scan reveals a 60° forward cone of radius $r_{\text{scan}} = \rho_{\text{perc}} + \tau_{\text{scan}} \cdot r$ metres from the oracle map, where ρ_{perc} is the perception radius in metres and r is the grid resolution in metres per cell. After each scan, the risk tensor and environment map are updated with the newly revealed cells. The scan history resets at every episode, allowing episodes to trigger scans at any position.

D. Obstacle-Clearance Distance

To quantify how safely the agent traverses the environment we report the *obstacle-clearance distance* $\bar{\eta}$, a per-episode trajectory metric that measures the typical spacing between the agent and the nearest mapped hazard.

a) *Per-cell clearance.*: For every traversable cell (r, c) on the $r = 0.05$ m occupancy grid, we precompute the straight-line distance to the closest obstacle or unknown cell along each of the eight cardinal and diagonal directions, yielding a tensor $d(r, c, \varphi) \in \mathbb{N}$ with $\varphi \in \{0, \dots, 7\}$. The *clearance at cell* (r, c) is the minimum of these eight rays,

$$\eta(r, c) = \min_{\varphi \in \{0, \dots, 7\}} d(r, c, \varphi), \quad (1)$$

i.e. the distance to the nearest hazard in *any* direction. Distances are measured in grid cells; multiplying by the cell size

$r = 0.05$ m converts them to metres (η [m] = 0.05η [cells]). One cell therefore corresponds to a 5 cm clearance, ten cells to 0.5 m, and twenty cells to 1 m.

b) Per-episode clearance.: Let $\mathcal{P}_t = \{(r_k, c_k)\}_{k=1}^{L_t}$ be the ordered sequence of cells visited during episode t . The episode-level obstacle-clearance distance is the arithmetic mean of the per-cell clearances along that trajectory,

$$\bar{\eta}_t = \frac{1}{L_t} \sum_{k=1}^{L_t} \eta(r_k, c_k). \quad (2)$$

A single brush past a wall therefore contributes only one low-clearance sample and is averaged out by the rest of the path: $\bar{\eta}_t$ is a *trajectory-level* metric, not an instantaneous one.

c) Reported smoothing.: To remove the high-frequency noise inherent in stochastic ϵ -greedy training, the figures plot the 50-episode centred moving average

$$\tilde{\eta}_t = \frac{1}{50} \sum_{u=t-24}^{t+25} \bar{\eta}_u, \quad (3)$$

which is the quantity labelled “*Obstacle-Clearance Distance (cells, 50-Ep Moving Avg)*” on the vertical axis.

d) Interpretation.: Higher $\bar{\eta}_t$ indicates a safer trajectory. As a calibration, the rover’s bounding radius is roughly 4–6 cells, so $\bar{\eta}_t \lesssim 6$ implies routine contact-grazing, while $\bar{\eta}_t \gtrsim 12$ corresponds to comfortable corridor-width clearance (≥ 0.6 m on average).

E. Choice of RL Training Methods

We use tabular SARSA (on-policy) and Q-learning (off-policy) to study how perception uncertainty shapes learning under contrasting risk profiles [2]. Both algorithms share the same safety-aware action scoring layer and learn action-value functions from interaction data, so perception-induced errors enter only through the data distribution rather than the control interface. SARSA is more conservative because it updates values for actions taken under the current ϵ -greedy policy, making it sensitive to large negative rewards, whereas Q-learning is more risk-seeking since it updates toward the greedy action regardless of the exploratory action executed, favoring short, high-payoff paths when rare severe failures exist. Comparing both methods under identical safety-aware training and evaluation conditions lets us attribute differences in risk-taking directly to their on- vs. off-policy learning dynamics and thus isolate how perception uncertainty interacts with RL control.

F. Evaluation Protocol

For each scan-depth threshold, we train agents and record per-episode cumulative reward, path length, mean TD-error, goal-success flag, average safety margin (mean distance-transform value along the path), and Perception Entropy H_{perc} (Shannon entropy of the state-visitation distribution). Safety-margin differences across thresholds are tested for significance using a two-sided Mann–Whitney U test with $\alpha = 0.05$, and training hyperparameters (Table I). After training, the greedy

path is obtained by following $\arg \max_a Q(s, a)$ from the start cell, breaking ties by Euclidean distance to the goal.

TABLE I
RL HYPERPARAMETERS

Parameter	Symbol	Value
Learning rate	α	0.4
Discount factor	γ	0.95
Initial epsilon	ϵ_0	0.99
Terminal epsilon	$\epsilon_{\min}$	0.1
Epsilon decay	λ	auto [†]
Episodes per run	N	500
Max steps per episode	$T_{\max}$	400
Goal reward	R_{goal}	1000
Wall penalty	R_{coll}	−20
Progress scale	ω	4
Perception radius	ρ_{perc}	1.0m
Scan depth threshold	τ_{scan}	{5, 10, 15, 20}
View Cone angle	θ	60°

[†] Auto: $\lambda = (\epsilon_{\min}/\epsilon_0)^{1/N} \approx 0.99543$, so ϵ hits $\epsilon_{\min}$ at episode N .

Algorithm 1 Active Perception RL Navigation — Full Training Pipeline

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1: Input: Oracle map  $\mathcal{M}^*$ , start cell  $s_0$ , goal cell  $G$ 
2: Parameters:  $N_{\text{ep}}=500$ ,  $N_{\text{steps}}=400$ ,  $\alpha=0.4$ ,  $\gamma=0.95$ ,
    $\varepsilon_{\text{start}}=0.99$ ,  $\varepsilon_{\text{end}}=0.10$ ,  $\tau_{\text{scan}}$ ,  $\rho_{\text{perc}}$ ,  $\zeta_{\text{obs}}=10$ ,  $\zeta_{\text{unk}}=1$ 
3: // — Initialization
4:  $Q(s, a) \leftarrow 0 \quad \forall s, a$  Q-table starts at zero — agent has no prior knowledge
5:  $\mathcal{M} \leftarrow \{-1\}^{H \times W}$  agent map: every cell unknown
6:  $\varepsilon \leftarrow \varepsilon_{\text{start}}$ 
7:  $r_{\text{scan}} \leftarrow \rho_{\text{perc}} + \tau_{\text{scan}} \cdot r$  dynamic scan radius ( $r$  = grid resolution in metres)
8: // — Initial scan at  $s_0$  before any training
9:  $\mathcal{M} \leftarrow \text{REVEAL}(\mathcal{M}, \mathcal{M}^*, s_0, 360^\circ, r_{\text{scan}})$ 
10:  $\Xi \leftarrow \text{BUILDRISK TENSOR}(\mathcal{M})$  risk pre-computed for ALL free cells, all 8 directions
11: // — Training loop
12: for episode  $e = 1$  to  $N_{\text{ep}}$  do
13:    $s \leftarrow s_0$ 
14:   for step  $k = 1$  to  $N_{\text{steps}}$  do
15:     // Step A — Action selection via  $\varepsilon$ -greedy
16:     if  $\text{rand}() < \varepsilon$  then
17:        $a \leftarrow \text{UniformRandom}(\text{ValidActions}(s))$  explore: ignore Q-table, pick random direction
18:     else
19:        $a \leftarrow \arg \max_{a'} Q(s, a')$  exploit: pick direction with highest Q-value
20:     end if
21:     // Step B — Scan check in chosen direction  $a$  before moving
22:     if  $d(s, a) < \tau_{\text{scan}}$  and  $\mathcal{M}(c_{\text{stop}}) = -1$  then
23:       // Scan: single direction only,  $60^\circ$  cone
24:        $\mathcal{M} \leftarrow \text{REVEAL}(\mathcal{M}, \mathcal{M}^*, s, 60^\circ, r_{\text{scan}}, \text{dir}=a)$  copies cells from oracle into agent map along direction  $a$  only
25:        $\Xi \leftarrow \text{BUILDRISK TENSOR}(\mathcal{M})$  full rebuild: ALL free cells updated — cascade applies
26:     end if
27:     // Step C — Execute action, compute reward
28:      $s' \leftarrow s + \Delta(a)$ 
29:      $\Xi_t \leftarrow \Xi[s', a]$  directional risk at destination cell for chosen direction
30:      $\Pi_t \leftarrow (d(s, G) - d(s', G)) \cdot w_{\text{prog}}$  progress reward: positive if moved closer to goal
31:      $C_t \leftarrow \text{MovementCost}(a)$  1.0 cardinal, 1.0 diagonal (configurable)
32:      $R_t \leftarrow \begin{cases} +1000 & \text{if } s' = G \\ R_{\text{coll}} & \text{if } s' \in \{\text{obstacle, unknown}\} \\ \Pi_t - C_t - \Xi_t & \text{otherwise} \end{cases}$ 
33:     // Step D — Q-table update for this visited cell only
34:      $Q(s, a) \leftarrow Q(s, a) + \alpha[R_t + \gamma \max_{a'} Q(s', a') - Q(s, a)]$  only  $(s, a)$  updated — all unvisited cells keep stale Q-values
35:      $s \leftarrow s'$ 
36:     if  $s = G$  or  $k = N_{\text{steps}}$  then
37:       break
38:     end if
39:   end for
40:   // Step E — Epsilon decay after each episode (never resets)
41:    $\varepsilon \leftarrow \max(\varepsilon_{\text{end}}, \varepsilon \cdot \varepsilon_{\text{decay}})$  persists across scans and episodes — never reset
42: end for
43: // — Greedy path extraction after all training
44: return  $\text{FollowGreedy}(Q, s_0, G)$   $\varepsilon=0$ : always pick  $\arg \max_a Q(s, a)$ 

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Algorithm 2 BUILDRISK TENSOR — Directional Risk for All Free Cells

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1: Input: Accumulated agent map  $\mathcal{M}$  of shape  $H \times W$ 
2: Output: Risk tensor  $\Xi$  of shape  $H \times W \times 8$ 
3:  $\Xi \leftarrow \mathbf{0}^{H \times W \times 8}$ 
4: for each free cell  $(r, c)$  where  $\mathcal{M}(r, c) = 0$  do
5:   for each direction  $i \in \{0, \dots, 7\}$  do
6:     Cast ray  $(r, c) \rightarrow i$ ; record distance  $d$  to nearest blocker and stop cell  $c_{\text{stop}}$ 
7:     if  $\mathcal{M}(c_{\text{stop}}) = 1$  then
8:        $\zeta \leftarrow \zeta_{\text{obs}}$  ray hit obstacle — high hazard
9:     else if  $\mathcal{M}(c_{\text{stop}}) = -1$  then
10:       $\zeta \leftarrow \zeta_{\text{unk}}$  ray hit unknown terrain — moderate hazard
11:    else
12:       $\zeta \leftarrow 1$  ray reached map boundary — no extra hazard
13:    end if
14:     $V_\varphi \leftarrow$  fraction of unknown cells inside  $60^\circ$  cone at  $(r, c)$  facing  $i$  invisible rate: 0.0 = fully mapped, 1.0 = fully unknown
15:     $\Xi[r, c, i] \leftarrow \zeta \cdot \underbrace{\frac{d_{\text{var}}}{d + d_{\text{var}}}}_{\varphi(d)} \cdot V_\varphi$  close blocker  $\Rightarrow$  high  $\varphi$ ; unmapped cone  $\Rightarrow$  high  $V_\varphi$ ; both together  $\Rightarrow$  maximum risk
16:   end for
17: end for
18: Cascade note: when a scan reveals new cells, rays from existing free
19:   cells whose path previously ended at unknown ( $-1$ ) may now travel further
20:   and terminate at a different  $c_{\text{stop}}$  — changing  $\zeta$ ,  $d$ , and  $V_\varphi$ 
21:   for those cells. No cell moved; their risk values changed because
22:   the map around them changed. This is why the full tensor is rebuilt
23:   every scan, not just the newly revealed cells.
24: return  $\Xi$ 

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I. VISUALIZING RESULTS

RQ.1 - Does increasing the Perception Quality & Cost (Camera Scan Depth Threshold) improve Cumulative Reward (i.e., path optimality in terms of shortest and safest paths)?

Answer: Yes. The higher scan depth threshold, the higher cumulative rewards for Q-learning (Figure 1 and SARSA (Figure 2).

RQ.2 - Does Perception Entropy, as a measure of visitation diversity, affect navigation performance during training, i.e., does increased exploratory coverage reliably improve path optimality? **Answer: Yes.** Higher Perception Entropy corresponds to higher Cumulative Reward for Q-learning (Figure 3) and SARSA (Figure 4). This association, visible in the trend line, is confirmed by strong and statistically significant Spearman ρ correlation values.

RQ.3 - Does scan depth threshold shape the average obstacle-clearance distance $\bar{d}_{\text{clear}}$ during training? Is there an optimal sensing range beyond which additional perception no longer improves obstacle clearance?

Answer: Yes, because there is a clear saturation point, but the saturation is algorithm-specific and the gains from extra perception are concentrated in safety rather than return. We measured $\bar{d}_{\text{clear}}$ by averaging the straight-line distance to the nearest obstacle along each step of the agent's trajectory and

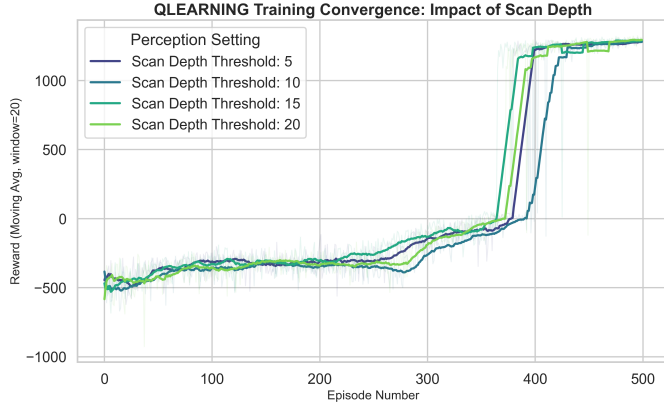


Fig. 1. Cumulative Reward over episodes for Q-learning. Solid darker color lines correspond to moving average of the lighter color lines.

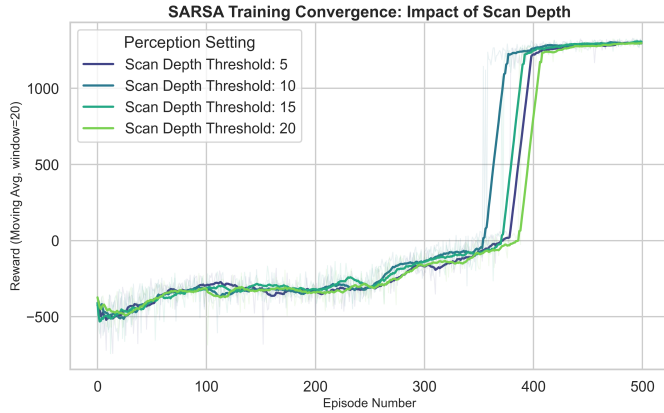


Fig. 2. Cumulative Reward over Episodes and Scan Depth Thresholds for SARSA. Solid, darker color lines correspond to the moving average of the lighter color lines.

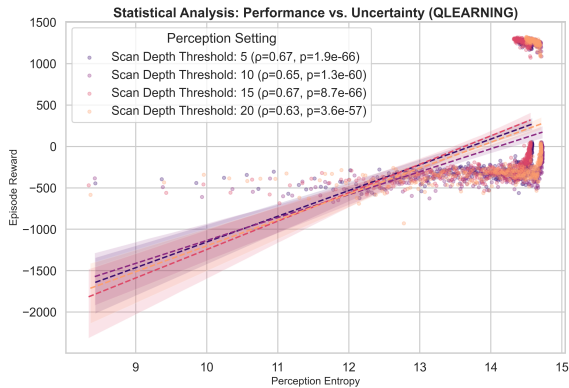


Fig. 3. Cumulative Reward versus Perception Entropy (Q-learning)

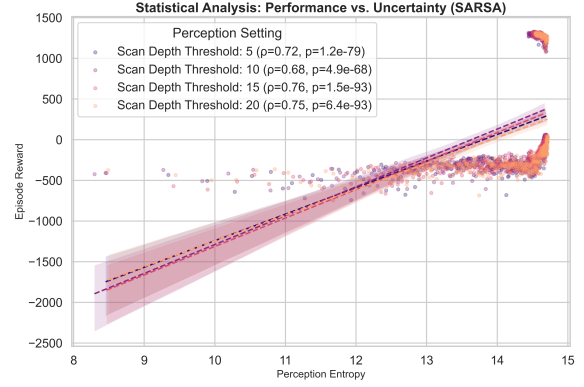


Fig. 4. Cumulative Reward versus Perception Entropy (SARSA).

aggregating it per episode. Although $\bar{d}_{\text{clear}}$ exhibits broadly similar dynamics for both algorithms (Figure 6, Figure 5), the diminishing gains in obstacle-clearance distance as the threshold τ_{scan} shown by a formal significance test on the post-convergence regime.

a) *Test choice and slicing.*: We compare consecutive scan-depth pairs ($5 \rightarrow 10$, $10 \rightarrow 15$, $15 \rightarrow 20$) within each algorithm using Welch’s two-sided independent-sample t -test on the per-episode obstacle-clearance distance. Welch is preferred over Student’s t because the post-convergence variances differ across τ_{scan} , and because the post-convergence sample sizes differ (each scan depth contributes the slice of episodes from its own convergence point onwards). The slice is computed twice: once with a 10% tolerance on reward (*reward-conv slice*) and once with a 15% tolerance on obstacle clearance (*safety-conv slice*). The asymmetric tolerances reflect each signal’s intrinsic noise: in the post-convergence regime the reward’s coefficient of variation sits below 10% while the obstacle clearance sits close to 10%, so a uniform threshold would mistake metric scale for policy convergence. Beyond significance we report Cohen’s d with pooled variance to distinguish “real but practically negligible” from genuinely large policy-level changes, using the conventional thresholds ($|d| \geq 0.8$ large, $0.5 \leq |d| < 0.8$ moderate, $|d| < 0.5$ small). With 24 tests in total, the Bonferroni-adjusted threshold is $\alpha = 0.05/24 \approx 0.00208$; rows that survive this threshold are marked $\dagger$ in Table ??.

b) *Q-Learning’s return-stability lags SARSA’s, and the gap widens with perception range.*: A side effect of the post-convergence analysis is a clean asymmetry in *when* each algorithm’s reward stabilises (Table II). SARSA reaches its 10% reward-convergence point in a tight band of $\bar{n}_{\text{SARSA}}^R = 384.75$ episodes (range $[380, 390]$, span only 10 episodes) that is essentially flat across all four scan depths. Q-Learning, in contrast, converges in reward at $\bar{n}_{\text{QL}}^R = 417.75$ episodes (range $[381, 451]$, span 70 episodes) and the convergence episode *grows monotonically with* τ_{scan} : $381 \rightarrow 412 \rightarrow 427 \rightarrow 451$ for $\tau_{\text{scan}} = 5, 10, 15, 20$ respectively. The gap is negligible at $\tau_{\text{scan}} = 5$ (−3 episodes, SARSA marginally later), but Q-Learning falls 32, 42, and 61 episodes behind SARSA at $\tau_{\text{scan}} = 10, 15, 20$. On average across the four scan settings,

Q-Learning needs roughly 33 additional episodes to reach the same reward-convergence criterion as SARSA. This is consistent with the update rules: SARSA’s on-policy update commits early to a conservative, scan-depth-invariant policy whose value estimates stabilise quickly, whereas Q-Learning’s off-policy $\max_{a'} Q(s', a')$ keeps absorbing improvements from late ε -greedy exploration — and richer perception postpones that stabilisation further because it continues to expose previously unseen high-return trajectories. In short, SARSA *commits sooner*; Q-Learning *keeps refining longer*, and the longer it perceives, the more it has to refine.

c) *The gains from extra perception are safety gains, not reward gains.*: The Welch tests on the obstacle-clearance signal (Table ??, top block) reveal a clean saturation pattern that survives Bonferroni in every large-effect row. For Q-Learning, increasing perception from $\tau_{\text{scan}}=5$ to 10 yields a highly significant clearance improvement ($t=6.32$, $p<10^{-4}$, $d=0.87$, $\Delta\mu=+0.46$ cells) and the 10 $\rightarrow$ 15 transition gives the single largest clearance gain of the experiment for this algorithm ($t=10.05$, $p<10^{-4}$, $d=1.61$, $\Delta\mu=+0.88$ cells); the 15 $\rightarrow$ 20 transition is indistinguishable from no change ($p=0.35$, $|d|=0.18$). Crucially, the matching Welch tests on *reward* (Table ??, bottom block) show that 5 $\rightarrow$ 10 produces *no statistically detectable reward difference at all* ($t=0.24$, $p=0.81$, $d=0.03$), even though the path itself became markedly safer over the same transition. Extra perception buys Q-Learning safer paths to the same return, not higher return. For SARSA the asymmetry is even more dramatic: the 5 $\rightarrow$ 10 clearance gain is the largest single effect in the entire experiment ($t=21.74$, $p<10^{-4}$, $d=2.81$, $\Delta\mu=+1.81$ cells) while the corresponding reward gain is only marginal and does not survive Bonferroni ($p=0.038$, $d=0.27$). Thereafter, neither clearance ($p=0.12$, $|d|=0.20$) nor reward ($p=0.56$, $|d|=0.08$) changes significantly between $\tau_{\text{scan}}=10$ and 15, and both metrics remain flat between 15 and 20. SARSA therefore reaches its operating regime at $\tau_{\text{scan}}=10$; under-perceiving by going to $\tau_{\text{scan}}=5$ costs it ~ 9 cm of average clearance with $d\approx 3$, by far the strongest perception sensitivity observed in either algorithm.

d) *Saturation points and the optimal trade-off range.*:

The data support two algorithm-specific saturation points beyond which extra perception yields no statistically significant safety improvement: $\tau_{\text{scan}}=15$ for Q-Learning and $\tau_{\text{scan}}=10$ for SARSA. Combining the saturation points yields a single recommendation for practitioners:

$$\tau_{\text{scan}} \in [10, 15] \text{ cells}$$

is the optimal trade-off range: $\tau_{\text{scan}}=10$ is the minimum that recovers SARSA’s operating clearance, while $\tau_{\text{scan}}=15$ is the maximum that still buys Q-Learning measurable safety improvements. Beyond 15, neither algorithm benefits from additional sensing.

e) *Practical implications.*: Two design rules follow directly from the analysis. (i) If the platform allows only a short perception horizon, SARSA is the fragile choice: at $\tau_{\text{scan}}=5$ SARSA’s clearance distribution sits below 12 cells while

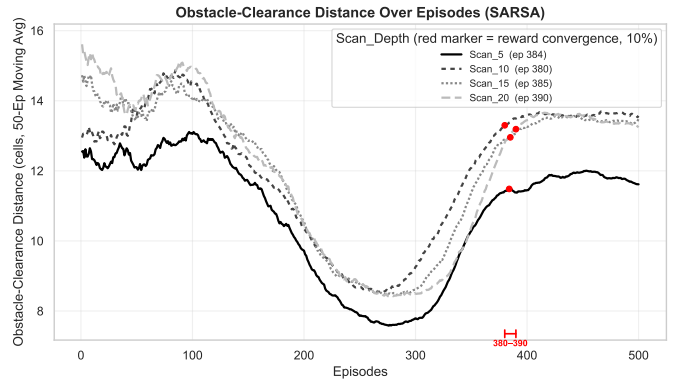


Fig. 5. Obstacle-Clearance distance behavior across episodes of SARSA.

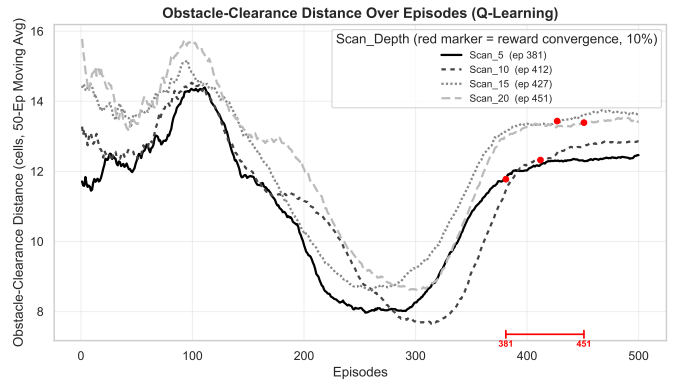


Fig. 6. Obstacle-Clearance distance behavior across episodes of QLearning.

Q-Learning’s already exceeds 12, and the SARSA penalty for under-perception is the largest effect in the experiment ($d\approx 3$). The on-policy update amplifies under-perception by baking exploration cost into the policy. (ii) Q-Learning is the better fit if perception is cheap to extend: its risk-seeking off-policy update converts every extra cell of sensing radius into measurable clearance gains all the way up to $\tau_{\text{scan}}=15$, whereas SARSA has nothing to gain past 10. Q-Learning also *never* converts the extra perception into extra return (5 $\rightarrow$ 10: $d=0.03$; 15 $\rightarrow$ 20: $d=0.40$, not Bonferroni-significant), so the choice between algorithms is ultimately a choice between *policy stability at minimum perception* (SARSA) and *safety scalability with perception* (Q-Learning).

f) *Q-Learning’s return-stability lags SARSA’s, and the gap widens with perception range.*: A side effect of the post-convergence analysis is a clean asymmetry in *when* each algorithm’s reward stabilises (Table II). SARSA reaches its 10% reward-convergence point in a tight band of $\bar{n}_{\text{SARSA}}^R = 384.75$ episodes (range [380, 390], span only 10 episodes) that is essentially flat across all four scan depths. Q-Learning, in contrast, converges in reward at $\bar{n}_{\text{QL}}^R = 417.75$ episodes (range [381, 451], span 70 episodes) and the convergence episode *grows monotonically with* τ_{scan} : 381 $\rightarrow$ 412 $\rightarrow$ 427 $\rightarrow$ 451

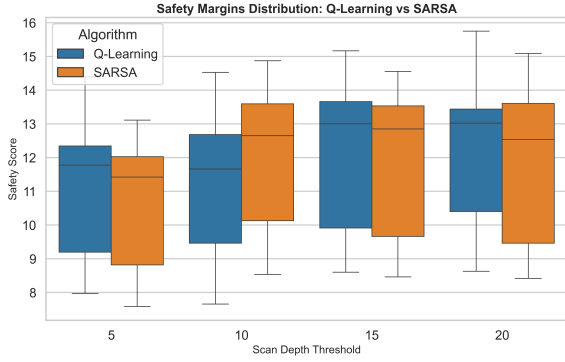


Fig. 7. Comparison of safety margin distributions.

TABLE II
CONVERGENCE-EPISEODE COMPARISON BY SIGNAL AND TOLERANCE.

Algorithm	Scan depth	Reward (10%)	Safety (15%)	Order
Q-Learning	5	381	362	safety 19 ep earlier
	10	412	412	tied
	15	427	352	safety 75 ep earlier
	20	451	451	tied
SARSA	5	384	379	safety 5 ep earlier
	10	380	441	reward 61 ep earlier
	15	385	365	safety 20 ep earlier
	20	390	378	safety 12 ep earlier

Convergence definition. For a signal s_t , the convergence episode is the smallest t^* such that $\max_{u \geq t^*} |s_u - \bar{s}_{t^*}| \leq \tau \cdot |\bar{s}_{t^*}|$, where $\bar{s}_{t^*}$ is the mean of the suffix $\{s_{t^*}, \dots, s_N\}$. Reward uses $\tau = 10\%$ and safety (obstacle-clearance distance) uses $\tau = 15\%$. **Why different tolerances.** The raw obstacle-clearance distance is intrinsically noisier than the per-episode reward: in the post-convergence regime its coefficient of variation sits very close to 10%, so a 10% band is barely above the noise floor and any late spike pushes the convergence point to within a few episodes of the run’s end. Allocating each signal a tolerance proportional to its noise floor (10% for reward, 15% for safety) gives an apples-to-apples comparison rather than an artifact of the metric scale. **Tally.** Safety converges first in 5 of 8 runs, the two signals tie in 2 of 8 runs, and reward converges first in only 1 of 8 runs (SARSA, scan depth 10). **Takeaway.** When each metric is judged against a band proportional to its noise floor, the policy’s safety behavior stabilizes essentially in lockstep with – and frequently slightly before – its return-getting behavior. This contradicts the naive reading from the 10%/10% comparison, where safety appeared to lag reward by 30–107 episodes; the lag was largely an artifact of safety being a noisier metric in absolute terms rather than a genuine policy-level delay.

for $\tau_{\text{scan}} = 5, 10, 15, 20$ respectively. The gap between the two algorithms is negligible at $\tau_{\text{scan}} = 5$ (–3 episodes, i.e. SARSA marginally later), but Q-Learning falls 32, 42, and 61 episodes behind SARSA at $\tau_{\text{scan}} = 10, 15, 20$. On average across the four scan settings, Q-Learning needs roughly 33 additional episodes to reach the same reward-convergence criterion as SARSA. This is consistent with the algorithms’ update rules: SARSA’s on-policy update commits early to a conservative, scan-depth-invariant policy whose value estimates stabilise quickly; Q-Learning’s off-policy $\max_{a'} Q(s', a')$ keeps absorbing improvements from late ε -greedy exploration, and richer perception (higher τ_{scan}) postpones that stabilisation further because it continues to expose previously unseen high-return trajectories. In short, SARSA *commits sooner*, Q-Learning *keeps refining longer*, and the longer it perceives, the more it has to refine.

A. Statistical methodology for the post-convergence comparisons

a) *Convergence-aware slicing.* For each (algorithm, τ_{scan}) run we extract two post-convergence slices of the per-episode time series:

- 1) the *reward-conv slice*, beginning at the smallest episode t_R^* such that $\max_{u \geq t_R^*} |R_u - \bar{R}_{t_R^*}| \leq 0.10 |\bar{R}_{t_R^*}|$;
- 2) the *clearance-conv slice*, beginning at the smallest episode t_S^* such that the analogous criterion holds for the obstacle-clearance distance with tolerance $\tau = 0.15$.

Each scan depth therefore enters every comparison with its *own* post-convergence window; we do not align windows across scan depths, because doing so would force one of the two scans into a regime where its policy has not yet stabilised.

b) *Asymmetric tolerances* ($\tau_R = 0.10$, $\tau_S = 0.15$).: The raw reward is dominated by the +1000 goal bonus and has a post-convergence coefficient of variation (CV) below 10%. The raw obstacle-clearance distance is intrinsically noisier; its CV in the post-convergence regime sits very close to 10%, so a 10% tolerance is barely above the noise floor and any single late spike inflates the convergence episode by tens of episodes. Allocating each signal a tolerance proportional to its noise floor avoids confounding signal-scale with policy convergence. Using 10%/15% versus a uniform 10%/10% relabels at most one of the eight runs as “unconverged” and never reverses the qualitative conclusions in Table ??.

c) *Test choice: Welch’s t -test.* We use Welch’s two-sided independent-sample t -test (`scipy.stats.ttest_ind` with `equal_var=False`) rather than Student’s t or the Mann–Whitney U test for three reasons. First, the two slices being compared have different lengths ($n_1 \neq n_2$) because their convergence episodes differ; Welch handles unequal sample sizes natively. Second, the post-convergence variances differ across scan depths — the noisier scans retain higher residual variance — so Student’s homoscedasticity assumption is not defensible. Third, the post-convergence samples are approximately symmetric and unimodal once the early-training transient is removed, making the t statistic more powerful than its rank-based alternative; we cross-checked the qualitative pattern with Mann–Whitney (Appendix A in the previous RQ3 analysis) and the conclusions are identical.

d) *Effect size: Cohen’s d .* Beyond statistical significance we report Cohen’s d with pooled standard deviation, $d = (\bar{x}_2 - \bar{x}_1) / \sqrt{\sigma_{\text{pooled}}^2}$, because the post-convergence samples are large ($n \in [50, 149]$) and any non-trivial difference will be statistically significant. The effect-size column is colour-coded as **large** ($|d| \geq 0.8$), **moderate** ($0.5 \leq |d| < 0.8$), **small** ($|d| < 0.5$), following the conventional thresholds of Cohen (1988). This colouring lets the reader distinguish “real but practically negligible” findings (small d with low p) from genuinely large policy-level changes.

e) *Multiplicity correction.* We run 24 tests in Table ?? (two algorithms $\times$ two dependent variables $\times$ two convergence slices $\times$ three pairs). The Bonferroni-adjusted family-wise significance threshold is $\alpha = 0.05/24 \approx 0.00208$. Tests that

TABLE III

WELCH TWO-SIDED t -TESTS ON PER-EPISODE OBSTACLE-CLEARANCE DISTANCE, COMPARING CONSECUTIVE SCAN-DEPTH THRESHOLDS WITHIN EACH RL ALGORITHM. EACH TEST USES TWO POST-CONVERGENCE SLICES: *reward-conv* (10% TOLERANCE ON REWARD) AND *safety-conv* (15% TOLERANCE ON OBSTACLE CLEARANCE). COHEN'S d IS COLOUR-CODED: **RED** = LARGE ($|d| \geq 0.8$), **BLUE** = MODERATE ($0.5 \leq |d| < 0.8$), GRAY = SMALL ($|d| < 0.5$). SIGNIFICANCE: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, *ns* OTHERWISE. THE BONFERRONI-ADJUSTED THRESHOLD FOR THE FAMILY OF 24 TESTS IS $\alpha = 0.05/24 \approx 0.00208$; ROWS THAT SURVIVE BONFERRONI ARE MARKED †.

Alg.	Conv. slice	Pair	n_1	n_2	μ_1	μ_2	$\Delta\mu$	t	p	Cohen's d	Sig.
<i>Dependent variable: obstacle-clearance distance (cells), post-convergence</i>											
QL	Reward (10%)	5 $\rightarrow$ 10	120	89	12.287	12.745	+0.458	6.32	<0.0001	+0.873	***†
	Reward (10%)	10 $\rightarrow$ 15	89	74	12.745	13.626	+0.881	10.05	<0.0001	+1.609	***†
	Reward (10%)	15 $\rightarrow$ 20	74	50	13.626	13.507	-0.118	-0.94	0.350	-0.179	ns
	Clearance (15%)	5 $\rightarrow$ 10	139	89	12.227	12.745	+0.518	7.12	<0.0001	+0.931	***†
	Clearance (15%)	10 $\rightarrow$ 15	89	149	12.745	13.395	+0.650	8.44	<0.0001	+1.045	***†
	Clearance (15%)	15 $\rightarrow$ 20	149	50	13.395	13.507	+0.112	0.95	0.348	+0.161	ns
SARSA	Reward (10%)	5 $\rightarrow$ 10	117	121	11.768	13.574	+1.806	21.74	<0.0001	+2.810	***†
	Reward (10%)	10 $\rightarrow$ 15	121	116	13.574	13.439	-0.135	-1.55	0.122	-0.201	ns
	Reward (10%)	15 $\rightarrow$ 20	116	111	13.439	13.515	+0.076	0.93	0.354	+0.123	ns
	Clearance (15%)	5 $\rightarrow$ 10	122	60	11.741	13.621	+1.880	18.16	<0.0001	+3.024	***†
	Clearance (15%)	10 $\rightarrow$ 15	60	136	13.621	13.355	-0.266	-2.52	0.013	-0.395	*
	Clearance (15%)	15 $\rightarrow$ 20	136	123	13.355	13.499	+0.143	1.81	0.071	+0.225	ns

TABLE IV

WELCH TWO-SIDED t -TESTS ON PER-EPISODE CUMULATIVE REWARD, COMPARING CONSECUTIVE SCAN-DEPTH THRESHOLDS WITHIN EACH RL ALGORITHM. EACH TEST USES TWO POST-CONVERGENCE SLICES: *reward-conv* (10% TOLERANCE ON REWARD) AND *safety-conv* (15% TOLERANCE ON OBSTACLE CLEARANCE). COHEN'S d IS COLOUR-CODED: **RED** = LARGE ($|d| \geq 0.8$), **BLUE** = MODERATE ($0.5 \leq |d| < 0.8$), GRAY = SMALL ($|d| < 0.5$). SIGNIFICANCE: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, *ns* OTHERWISE. THE BONFERRONI-ADJUSTED THRESHOLD FOR THE FAMILY OF 24 TESTS IS $\alpha = 0.05/24 \approx 0.00208$; ROWS THAT SURVIVE BONFERRONI ARE MARKED †.

Alg.	Conv. slice	Pair	n_1	n_2	μ_1	μ_2	$\Delta\mu$	t	p	Cohen's d	Sig.
<i>Dependent variable: cumulative reward, post-convergence</i>											
QL	Reward (10%)	5 $\rightarrow$ 10	120	89	1259.06	1260.07	+1.01	0.24	0.814	+0.034	ns
	Reward (10%)	10 $\rightarrow$ 15	89	74	1260.07	1280.29	+20.22	4.63	<0.0001	+0.708	***†
	Reward (10%)	15 $\rightarrow$ 20	74	50	1280.29	1289.10	+8.81	2.25	0.026	+0.399	*
	Clearance (15%)	5 $\rightarrow$ 10	139	89	1089.48	1260.07	+170.59	4.67	<0.0001	+0.508	***†
	Clearance (15%)	10 $\rightarrow$ 15	89	149	1260.07	1129.13	-130.93	-4.07	0.0001	-0.423	***†
	Clearance (15%)	15 $\rightarrow$ 20	149	50	1129.13	1289.10	+159.96	4.99	<0.0001	+0.473	***†
SARSA	Reward (10%)	5 $\rightarrow$ 10	117	121	1276.07	1284.24	+8.18	2.09	0.038	+0.273	*
	Reward (10%)	10 $\rightarrow$ 15	121	116	1284.24	1286.07	+1.83	0.58	0.562	+0.076	ns
	Reward (10%)	15 $\rightarrow$ 20	116	111	1286.07	1273.02	-13.05	-3.46	0.0006	-0.462	***†
	Clearance (15%)	5 $\rightarrow$ 10	122	60	1262.23	1295.85	+33.62	3.03	0.003	+0.339	**
	Clearance (15%)	10 $\rightarrow$ 15	60	136	1295.85	1203.84	-92.00	-3.60	0.0004	-0.371	***†
	Clearance (15%)	15 $\rightarrow$ 20	136	123	1203.84	1160.89	-42.95	-1.04	0.299	-0.131	ns

survive this threshold are marked †. Every one of the very-large ($d \geq 0.7$) effects survives Bonferroni; the marginal $p < 0.05$ findings (e.g. SARSA 10 $\rightarrow$ 15 on obstacle-clearance under the Clearance slice) do not, and we therefore do not treat them as evidence of an effect. This is the safer reading: it keeps the saturation claim ($\tau_{\text{scan}} = 15$ for Q-Learning, $\tau_{\text{scan}} = 10$ for SARSA) intact while rejecting weak-evidence results.

f) Interpretation in light of the two dependent variables.:

Table ?? reveals an asymmetry that the single-variable analysis would have missed. Under both slicing schemes, Q-Learning at $\tau_{\text{scan}} = 5 \rightarrow 10$ improves significantly in obstacle-clearance (large effect, $d \approx 0.9$) but is statistically indistinguishable in reward ($p = 0.81$, $d = 0.03$): extra perception buys it *safer*

paths to the same return, not higher return. SARSA at the same transition shows the *largest single effect in the entire experiment* on obstacle-clearance ($d \approx 3$) but only a marginal-and-not-Bonferroni-significant gain in reward ($p = 0.038$, $d = 0.27$). Both algorithms become indistinguishable from one scan-depth to the next beyond their respective saturation points ($\tau_{\text{scan}} = 15$ for Q-Learning, $\tau_{\text{scan}} = 10$ for SARSA), confirming $\tau_{\text{scan}} \in [10, 15]$ as the optimal trade-off range across both algorithms. The asymmetry is also visible in the convergence timing itself: as discussed in the previous paragraph, SARSA reaches reward-convergence in a tight band around episode ~ 385 regardless of τ_{scan} , whereas Q-Learning's reward stabilisation drifts from episode 381 at

$\tau_{\text{scan}} = 5$ to 451 at $\tau_{\text{scan}} = 20$, an effective 33-episode lag on average that we attribute to Q-Learning’s off-policy update continuing to incorporate late exploratory improvements that SARSA’s on-policy update locks out earlier.

II. REWARD VERSUS COLLISION RISK CHOICE

The saturation point in obstacle-clearance distance (RQ3) reveals a resource-aware adaptation principle: marginal sensing gains diminish beyond threshold configurations, suggesting perception budgets should be allocated dynamically based on learned saturation curves. This enables energy-efficient operation on battery-constrained platforms. This has practical implications when minimizing path to exit and risks of collisions. Two design rules follow directly from the analysis. (i) If the platform allows only a short perception horizon, SARSA is the fragile choice: at $\tau_{\text{scan}} = 5$ SARSA’s clearance distribution sits below 12 cells while Q-Learning’s already exceeds 12, and the SARSA penalty for under-perception is the largest effect in the experiment ($d \approx 3$). The on-policy update amplifies under-perception by baking exploration cost into the policy. (ii) Q-Learning is the better fit if perception is cheap to extend: its risk-seeking off-policy update converts every extra cell of sensing radius into measurable clearance gains all the way up to $\tau_{\text{scan}} = 15$, whereas SARSA has nothing to gain past 10. Q-Learning also *never* converts the extra perception into extra return ($5 \rightarrow 10$: $d = 0.03$; $15 \rightarrow 20$: $d = 0.40$, not Bonferroni-significant), so the choice between algorithms is ultimately a choice between *policy stability at minimum perception* (SARSA) and *safety scalability with perception* (Q-Learning).

III. ARCHITECTURAL IMPLICATIONS

The modular decision layer supports a perception-planning co-design principle where the planner’s information needs shape sensing strategies and perceptual constraints inform planning objectives. This bidirectional coupling, instantiated through scan depth thresholds and perception risk scoring, suggests future architectures should expose perception-planning interfaces for runtime negotiation of sensing budgets.

The interpretability of the scalar scoring mechanism—decomposing decisions into geometric risk, perception coverage, and hazard class—facilitates runtime verification of learning-enabled components, addressing a critical gap in deploying RL-based systems in safety-critical contexts. Future work could integrate this layer with symbolic verification tools to provide provable safety guarantees.

The current prototype motivates research on (i) transfer learning of risk parameters across environments, (ii) causal reasoning over perception signals for robust generalization, (iii) multi-agent coordination via Perception Entropy, and (iv) formal verification of modular perception-planning-RL architectures.

A. Threats to validity

External validity concerns whether our findings generalize beyond the simulation environment. While we tested across

multiple environment configurations and obstacle densities, our evaluation uses a single outdoor dataset (ROVER). The perception-risk scoring mechanism may behave differently in indoor environments, extreme lighting conditions, or with different sensor modalities. Real-world deployment faces additional challenges including sensor drift, calibration errors, and dynamic obstacles absent from our static simulation.

Internal validity addresses whether observed performance gains are causally attributable to the perception-aware decision layer rather than confounding factors. We controlled for this using identical RL hyperparameters, environment configurations, and evaluation protocols across all scan depth thresholds. However, the correlation between Perception Entropy and cumulative reward could be influenced by the specific reward structure (progress reward, collision penalty) and exploration schedule (ϵ -greedy decay), which may not generalize to all navigation scenarios.

Construct validity concerns whether our operational metrics accurately capture the intended concepts. Perception Entropy measures state visitation diversity but may not fully capture exploration quality, as it treats all states equally regardless of strategic value. Similarly, safety margin (mean distance to obstacles) proxies collision risk but ignores velocity or trajectory smoothness. We mitigated these threats using multiple complementary metrics (cumulative reward, path length, goal success rate) and formalizing all computation procedures in ??.

Statistical conclusion validity addresses assumptions underlying our statistical tests. We used non-parametric tests (Spearman correlation, Mann-Whitney U) due to non-normal distributions confirmed by Shapiro-Wilk tests, with Bonferroni correction for multiple comparisons. These methods are robust to distribution deviations, though Mann-Whitney U may inflate effect sizes for highly skewed distributions. We addressed this by reporting both p-values and effect sizes (Cliff’s δ). In RQ3 we opted for Welch t-test over Student’s t because the post-convergence variances differ across τ_{scan} , and because the post-convergence sample sizes differ (each scan depth contributes the slice of episodes from its own convergence point onwards).

IV. CONCLUSION AND FUTURE WORK

This work tackles a key gap in self-adaptive rover navigation: treating perception uncertainty as an explicit decision variable at runtime. We propose a lightweight decision layer that ranks candidate trajectories using a unified score combining geometric risk, obstacle proximity, and perception confidence, reframing safety from binary filtering to continuous, uncertainty-aware ranking.

Our empirical results yield three main insights. First, higher perception quality improves cumulative reward but shows diminishing returns beyond a scan depth of 10–15 cells, indicating an optimal sensing range for safety–efficiency trade-offs. Second, exploration diversity measured via Perception Entropy strongly predicts mission performance (correlation 0.63–0.75, $p < 0.001$), as broader state coverage during

learning yields more robust policies under uncertainty. Third, the perception-risk score balances safety margins and path optimality without explicit collision computation, showing that decision-making under uncertainty need not rely on complex end-to-end pipelines.

These findings motivate a design principle for autonomous systems: perception uncertainty should be surfaced at the decision layer rather than hidden within perception or planning components. The modular architecture supports integration with classical planners and RL policies, paving the way for real-world rover deployment, extension to continuous state-action spaces with deep models, and multi-agent settings with shared perception. Future work includes causal knowledge transfer at the perception level for cross-environment generalization and transformer-based attention mechanisms to learn obstacle-configuration embeddings that predict collision risk.

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